روش‌های پیشرفته نمونه برداري در جنگل

نمونه برداري فاصله اي
Distance sampling

هرمز سهرابي، دانشکده منابع طبيعي دانشگاه تربيت مدرس

مباني نمونه برداري فاصله‌اي

- ✓ فاصله نقطه تا فرد (Point to individual)
- ✓ فاصله فرد تا فرد (Individual to individual)

مدل مفهومي نمونه برداري

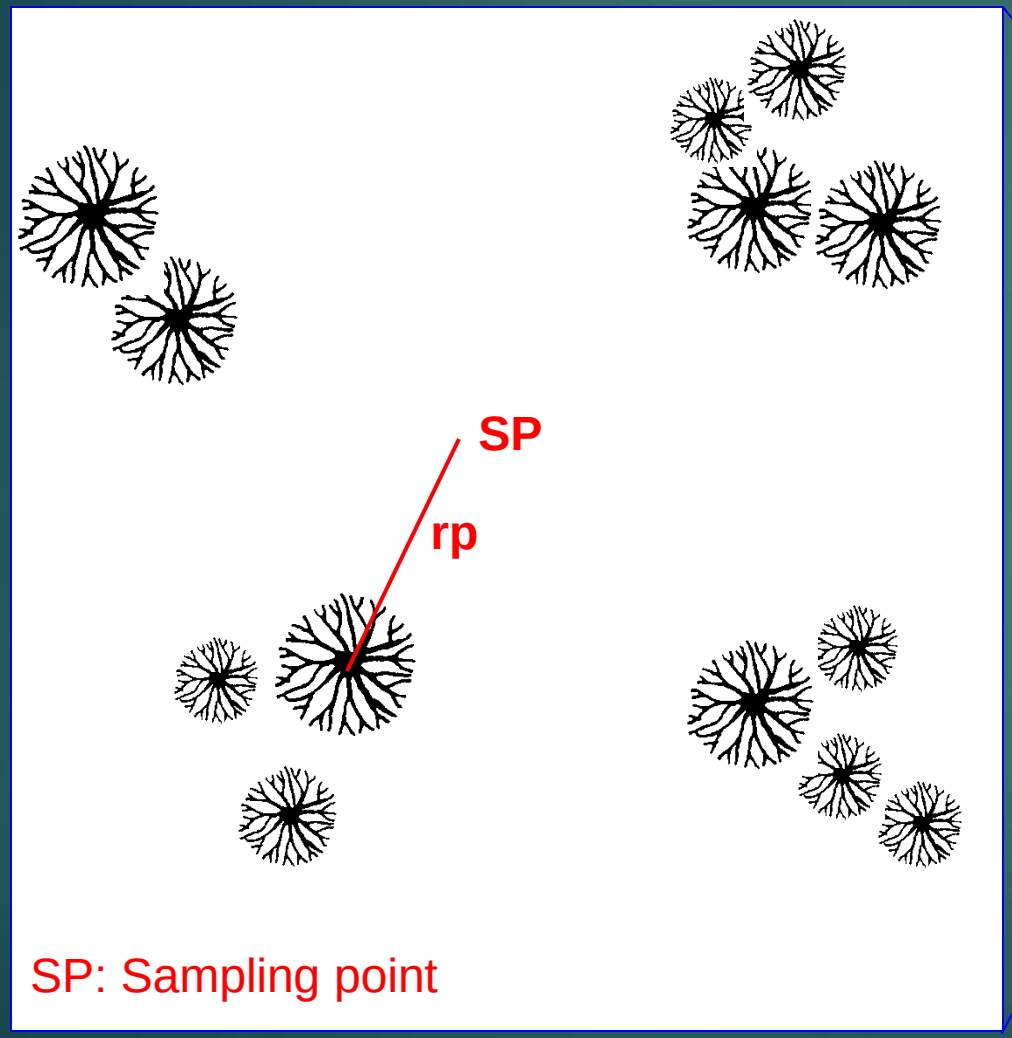
- Protocol
- طراحي نمونه برداري Sampling design
  - روش نمونه برداري Sampling Method
    - نمونه برداري نقطه‌اي Point Sampling
    - نمونه برداري فاصله‌اي Distance Sampling
    - نمونه برداري خطي Line Sampling
    - نمونه برداري قطعه‌اي Plot Sampling
  - الگوي نمونه برداري Sampling patterns
    - تصادفي ساده Simple Random S.
    - بلوك تصادفي Block Random S.
    - سيستماتيک تصادفی ساده Systematic S.
    - سيستماتيک مبتنی بر خطوط ميزان (Contour sampling)
  - اجتماع نمونه Sampling clusering
    - نمونه واحد Single s.
    - نمونه خوشه‌اي Cluster S.
  - استراتژي نمونه برداري Sampling strategies
    - مونه بندي
      - پيشا مونه بندي(Prestratification)
      - پسا مونه بندي (Poststratification)
    - نمونه برداري چند گانه Multiphase S.
      - Two phase sampling with regression estimator
      - Two phase sampling with ratio estimator
      - Two phase sampling for stratification
      - .....
    - نمونه برداري چند گامي Multistage S.
      - Two stage sampling
      - Three stage sampling
      - .....

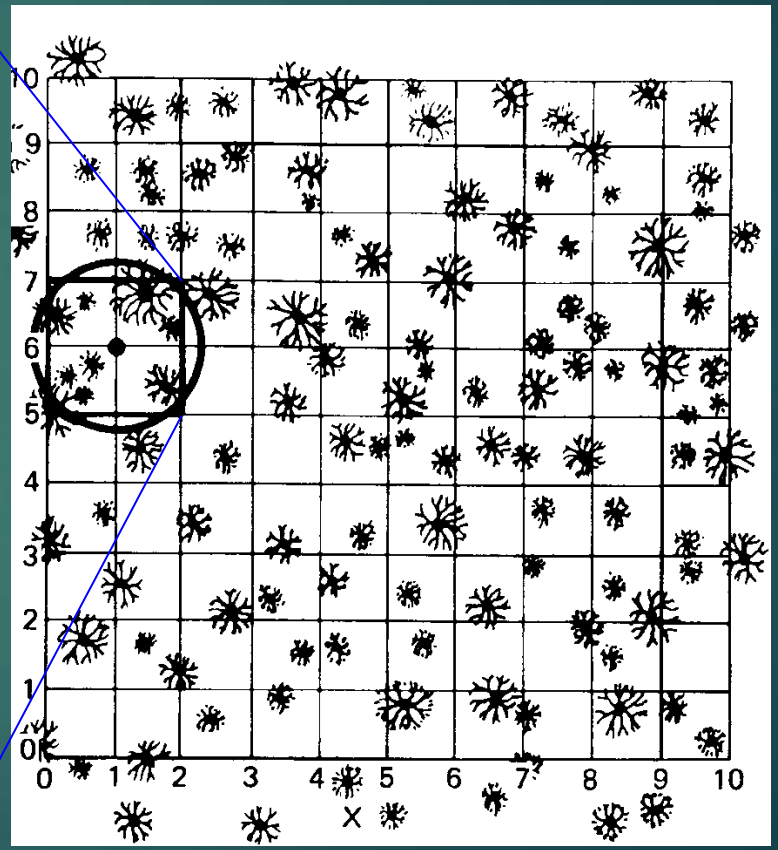
روش‌هاي رايج نمونه برداري فاصله‌اي

- نزديك‌ترين فرد
- Nearest Individual
- نزديك‌ترين همسايه
- Nearest Neighbor
- دومين نزديك‌ترين همسايه
- Second Nearest Neighbor
- روش مركب
- Compound Method
- نقاط متصل
- Joint-Point

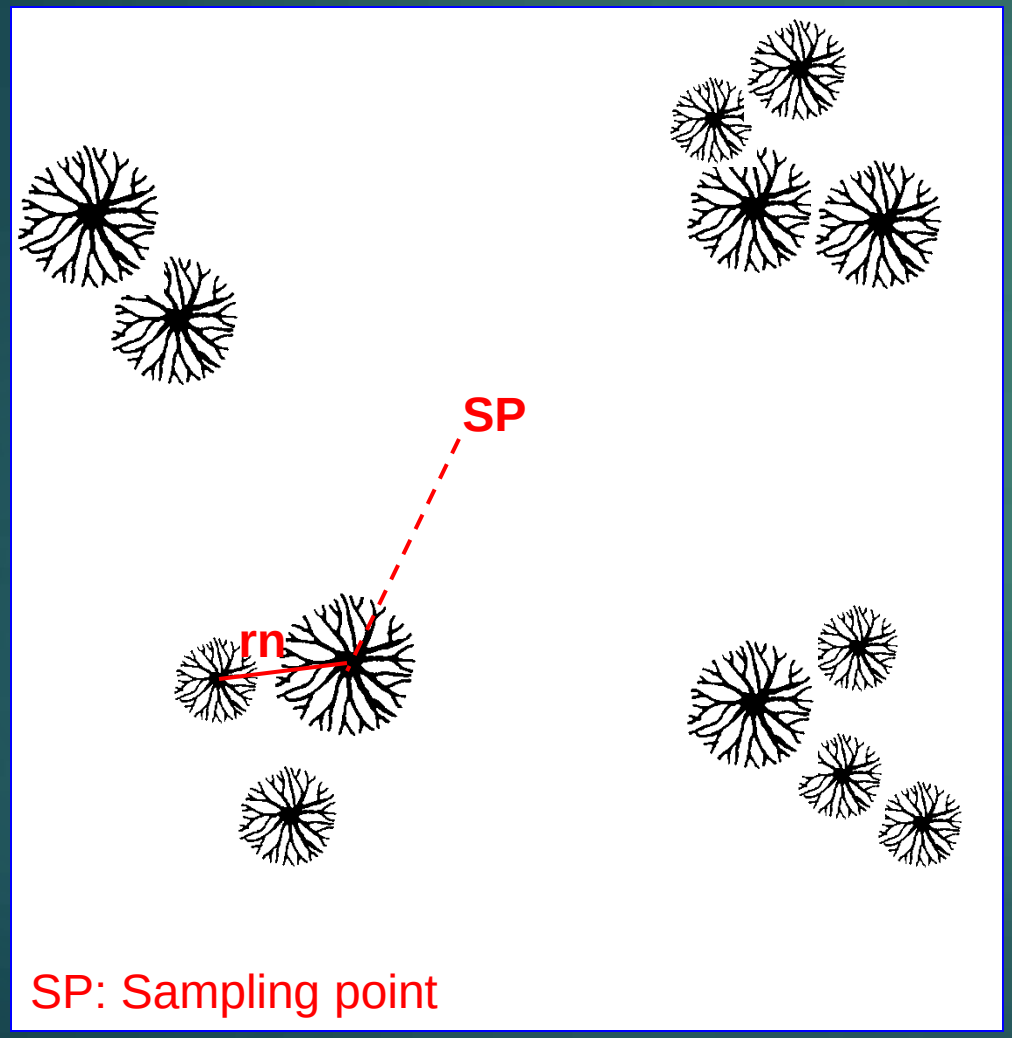
- زوج‌هاي تصادفي
- Random Pairs
- مربع تي
- T-Square
- فاصله مرتب
- Ordered Distance
- چارك با نقطه مركزي
- Point Centered Quarter
- چارك سرگردان
- Wandering Quarter

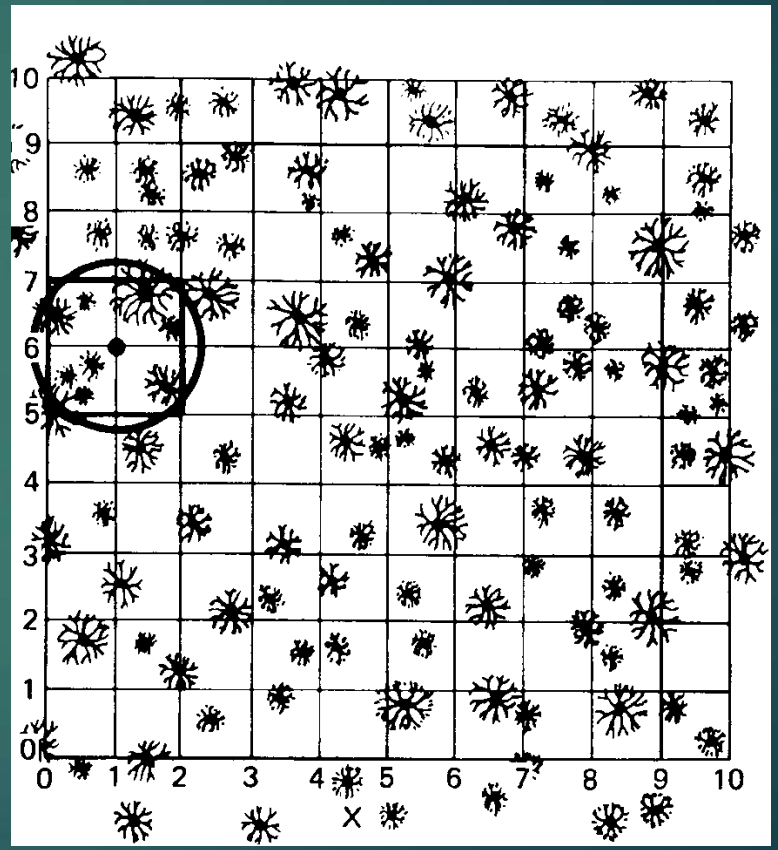
نزدیك‌ترین فرد (Nearest Individual)



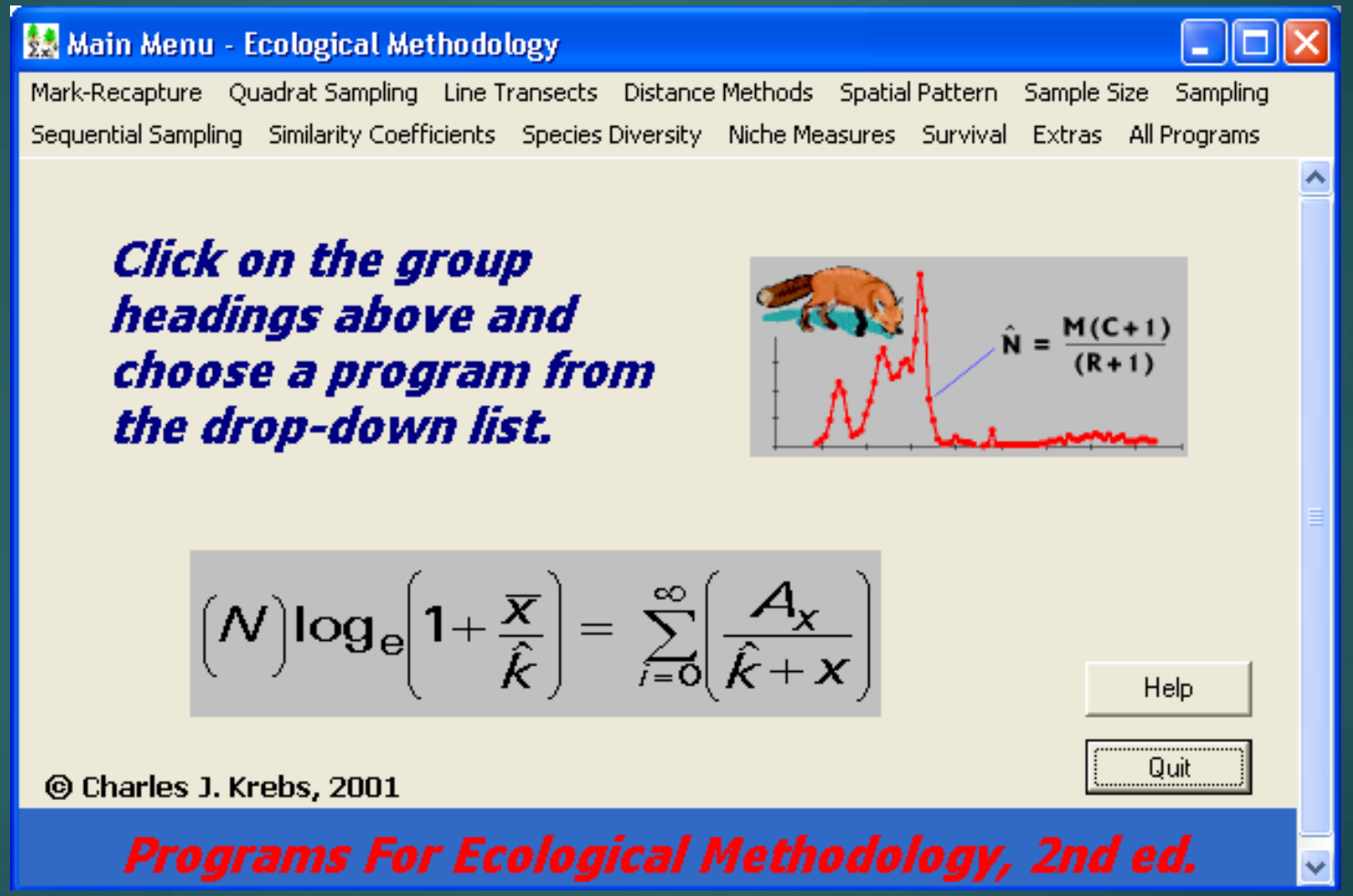


نزدیك‌ترین همسایه (Nearest Neighbor)

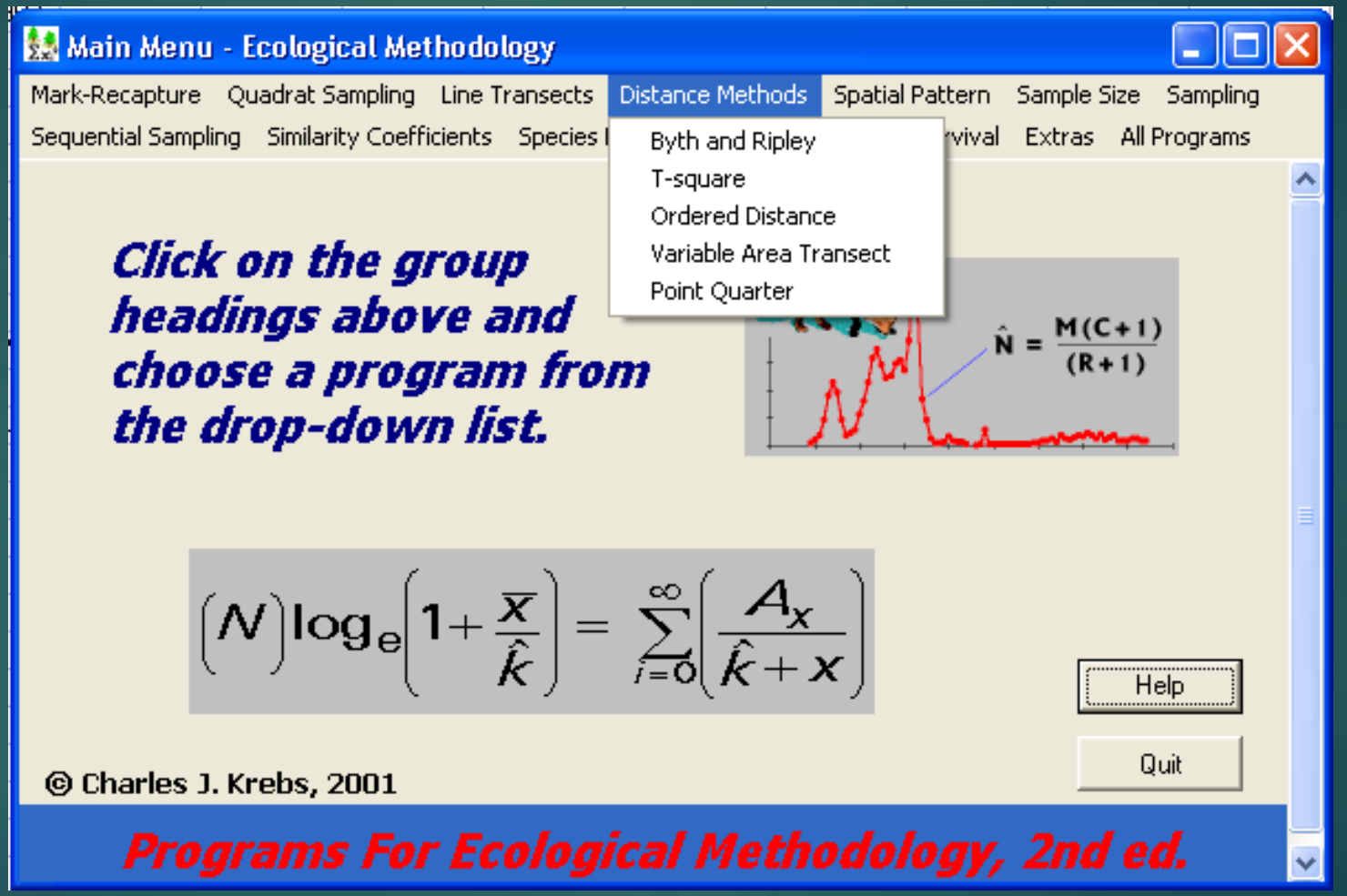




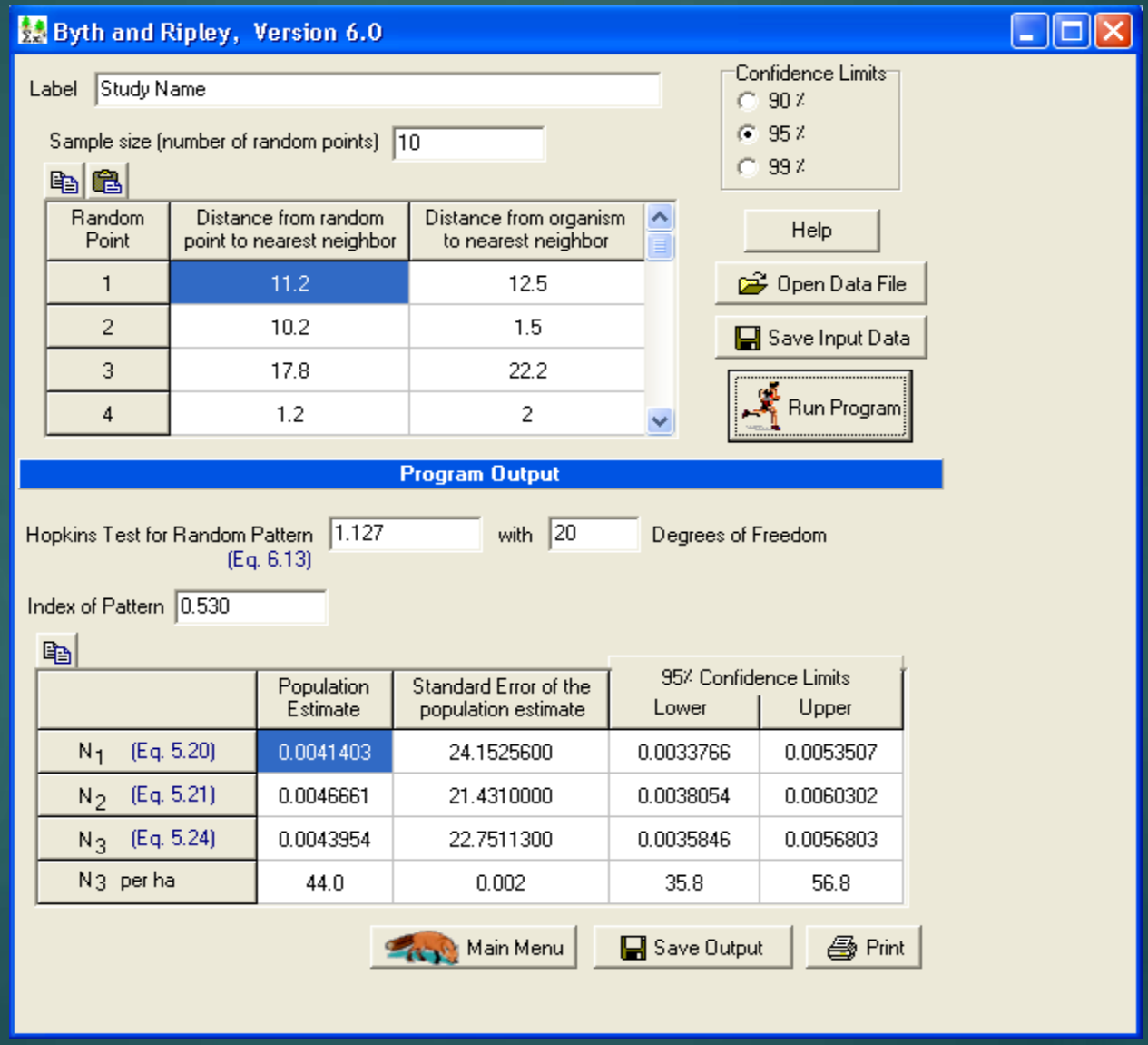
Ecological Methodology Software (2001, Ver6.0)



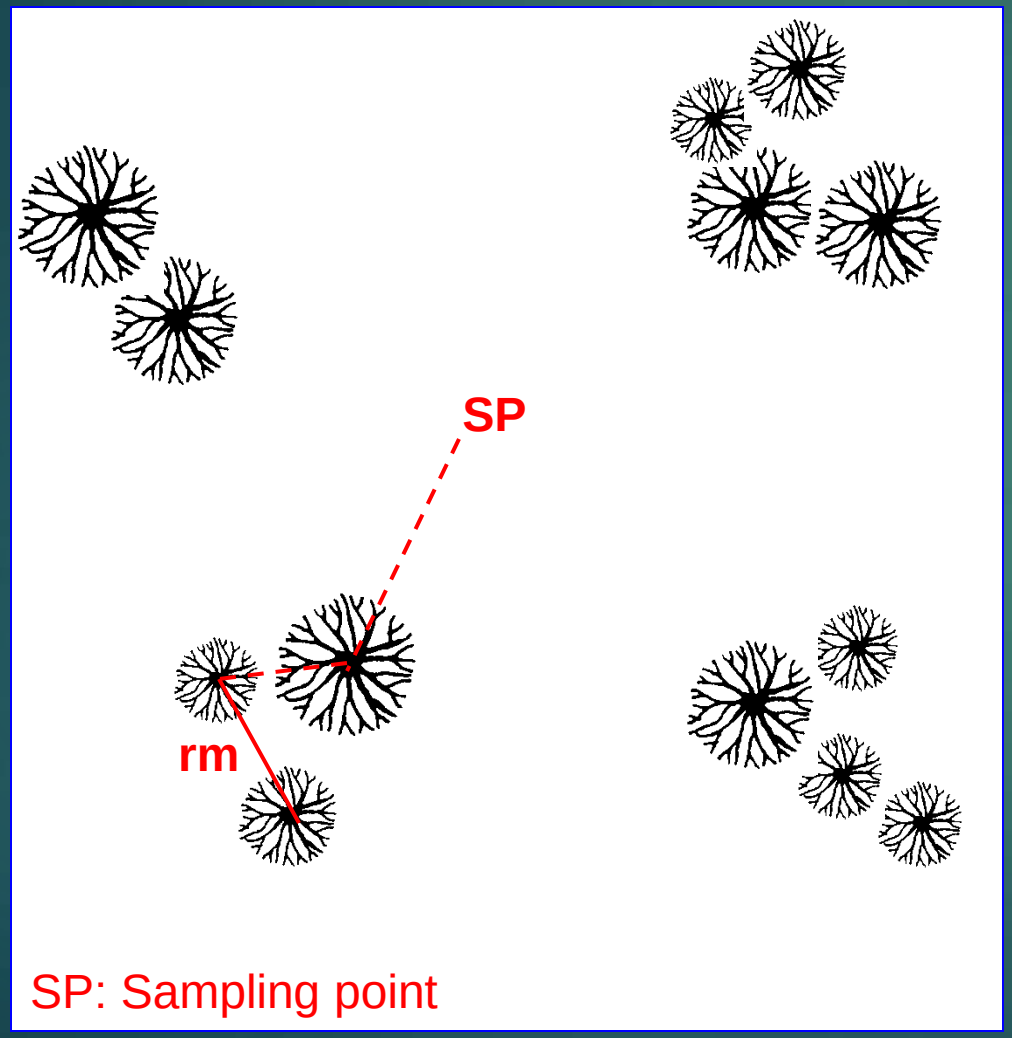
Ecological Methodology Software (2001, Ver6.0)

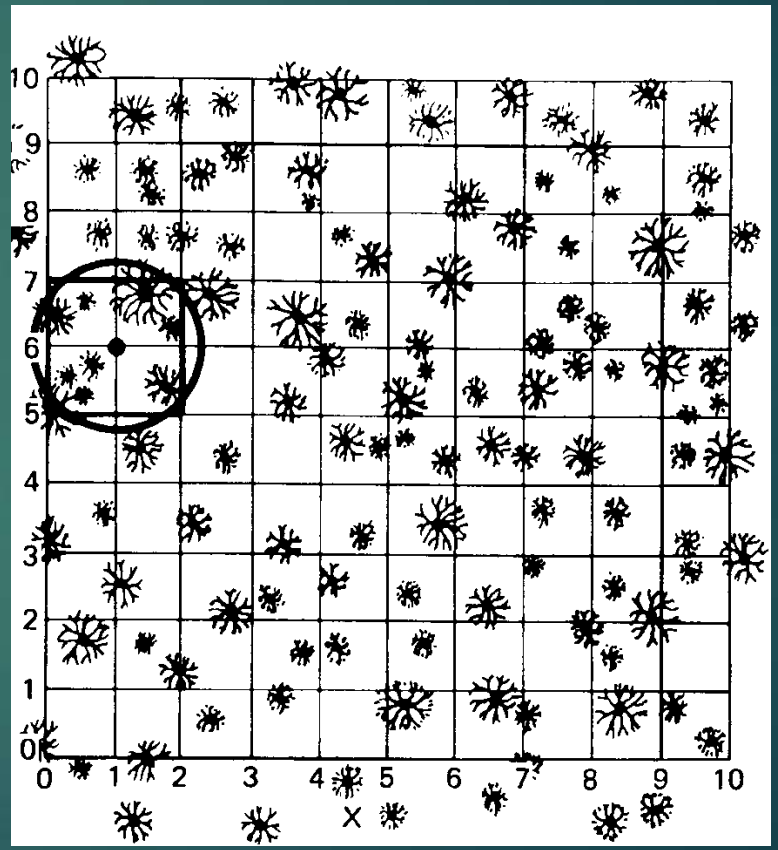


Ecological Methodology Software

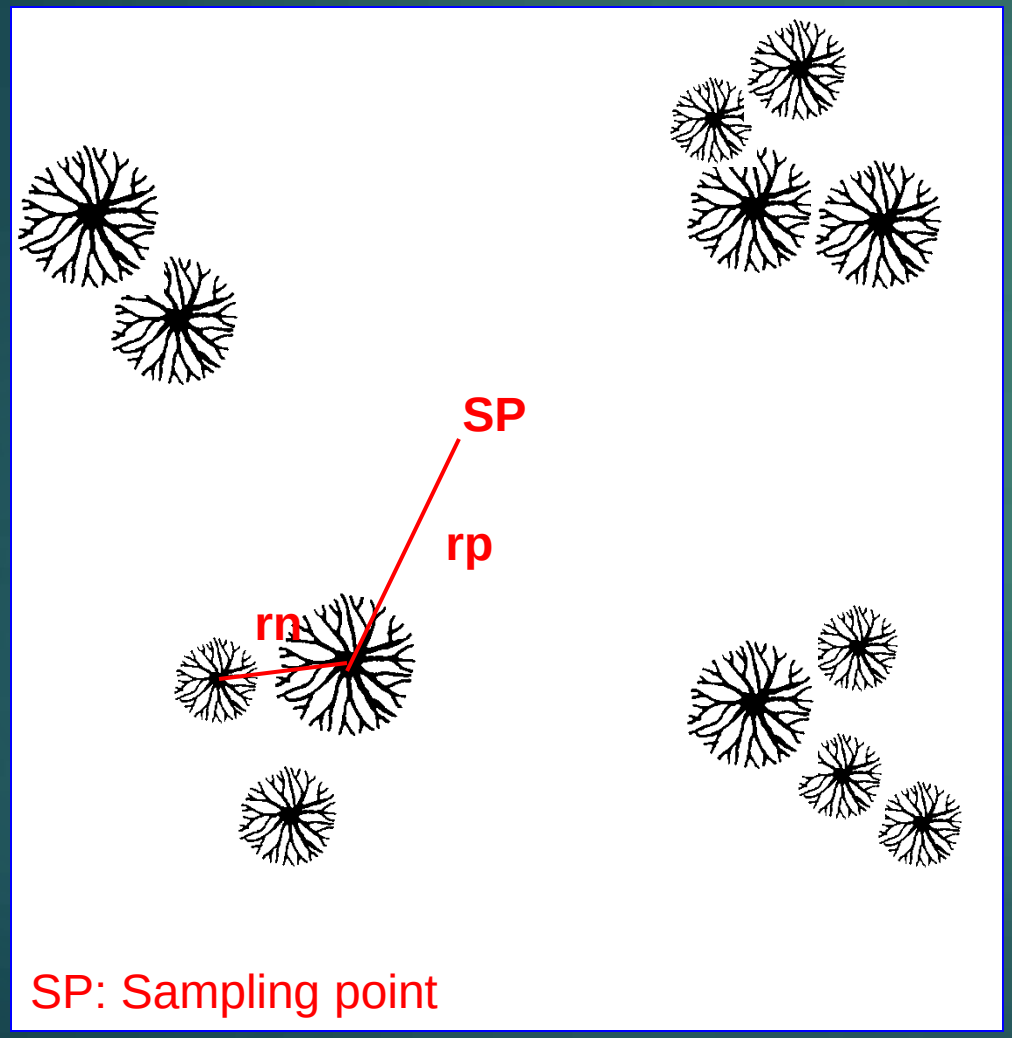


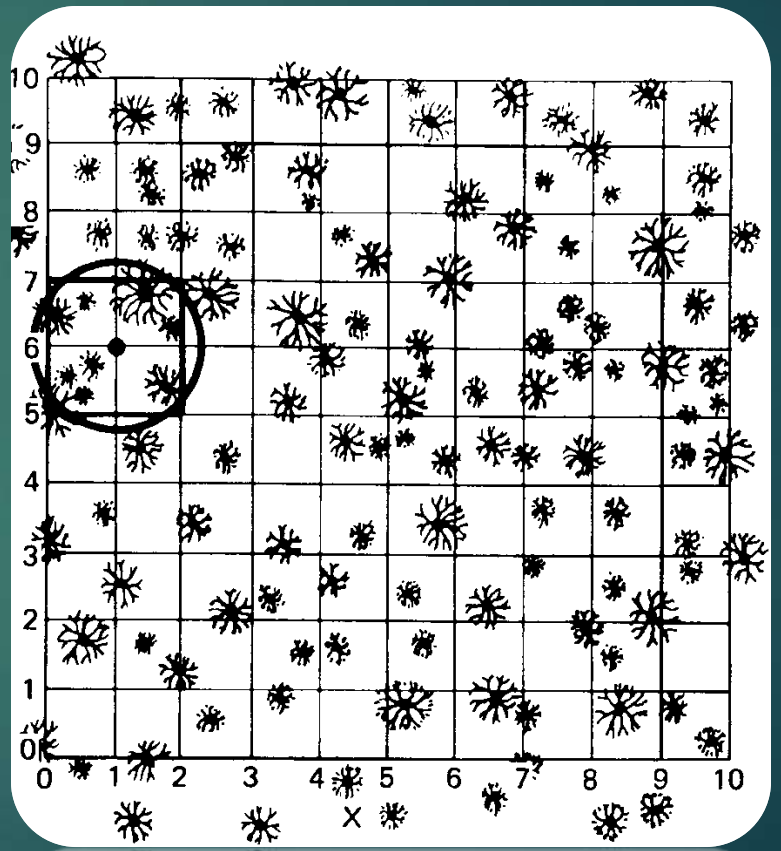
دومین نزدیك‌ترین همسایه (Second Nearest Neighbor)



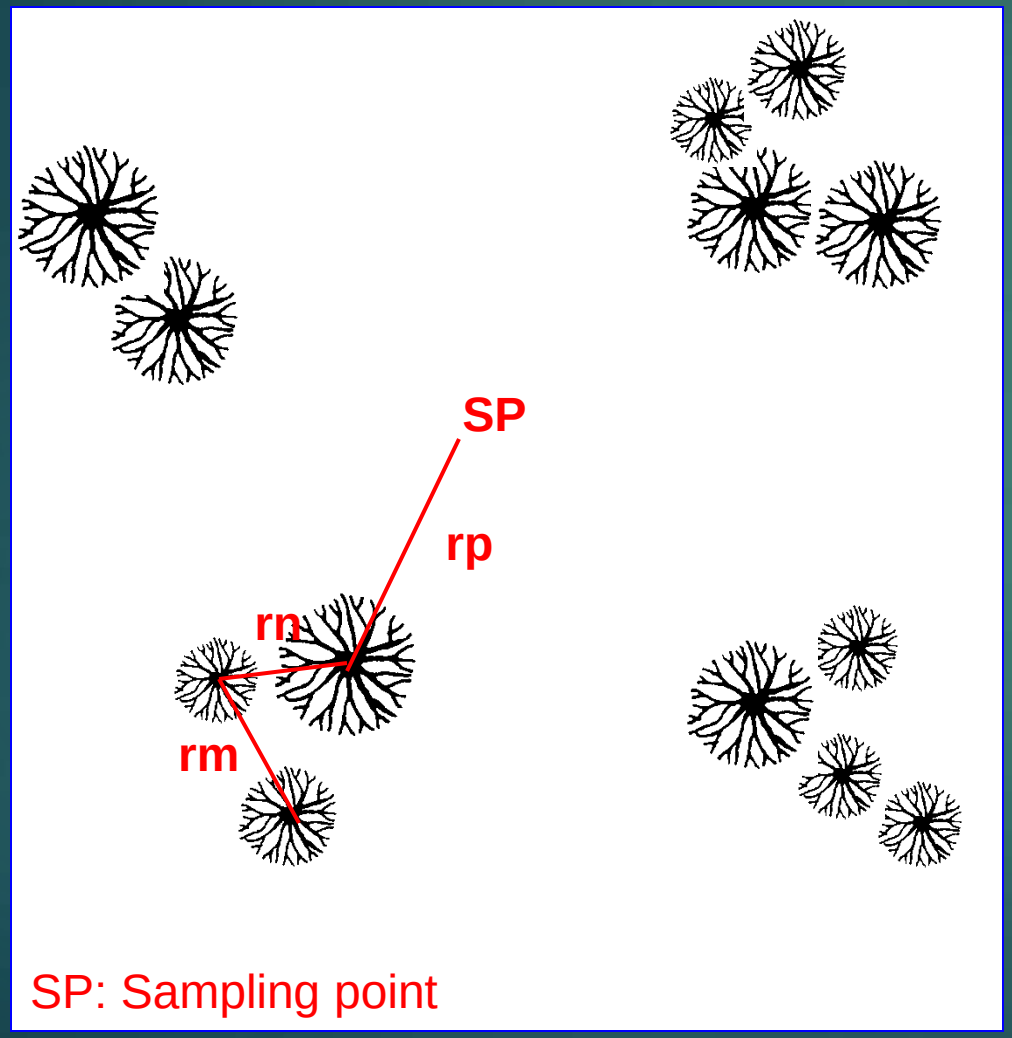


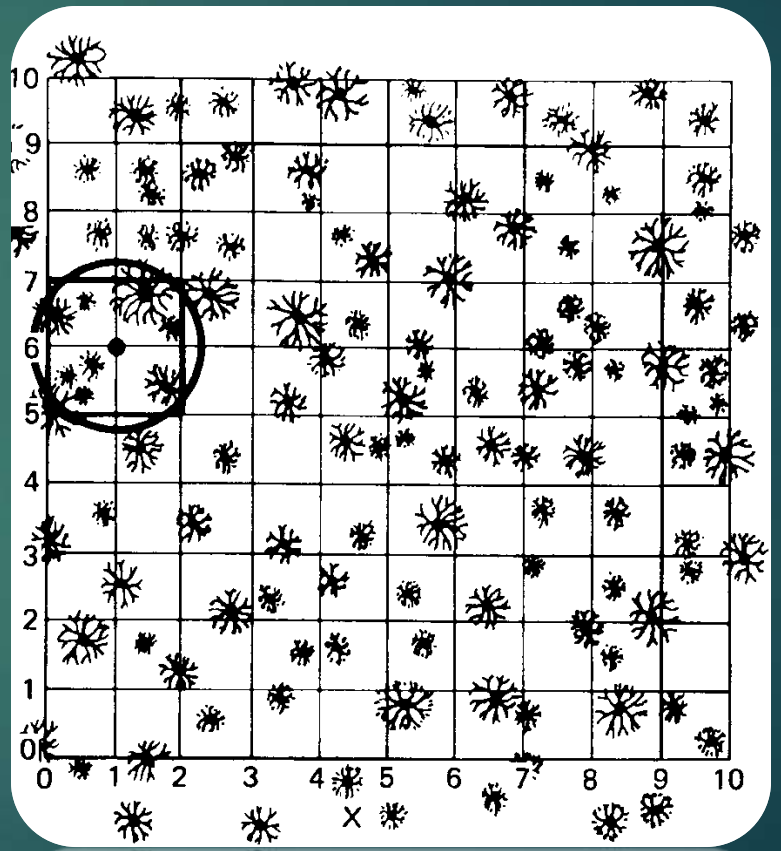
روش مرکب (Compound Method)



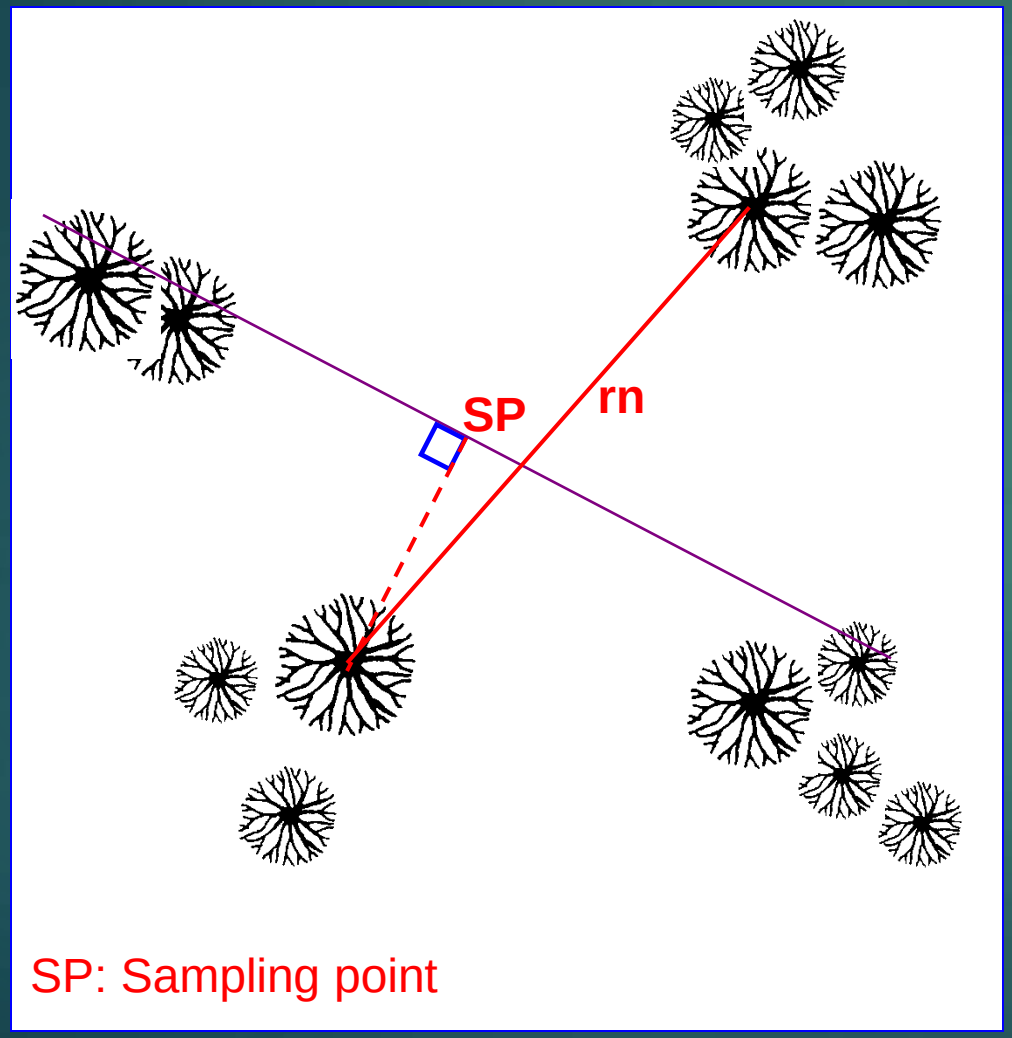


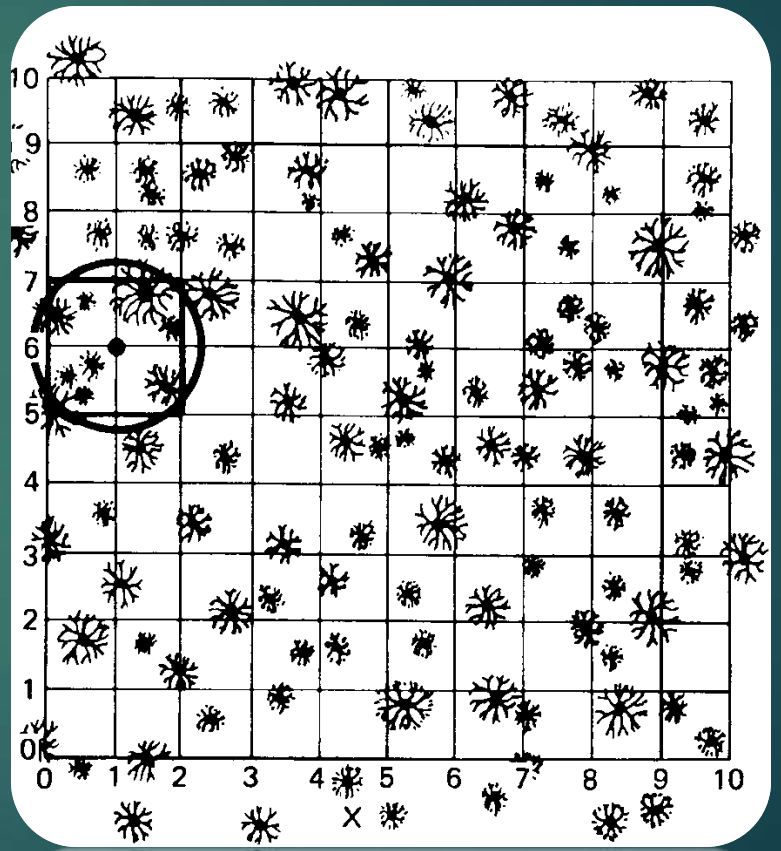
نقاط متصل (Joint Point Method)



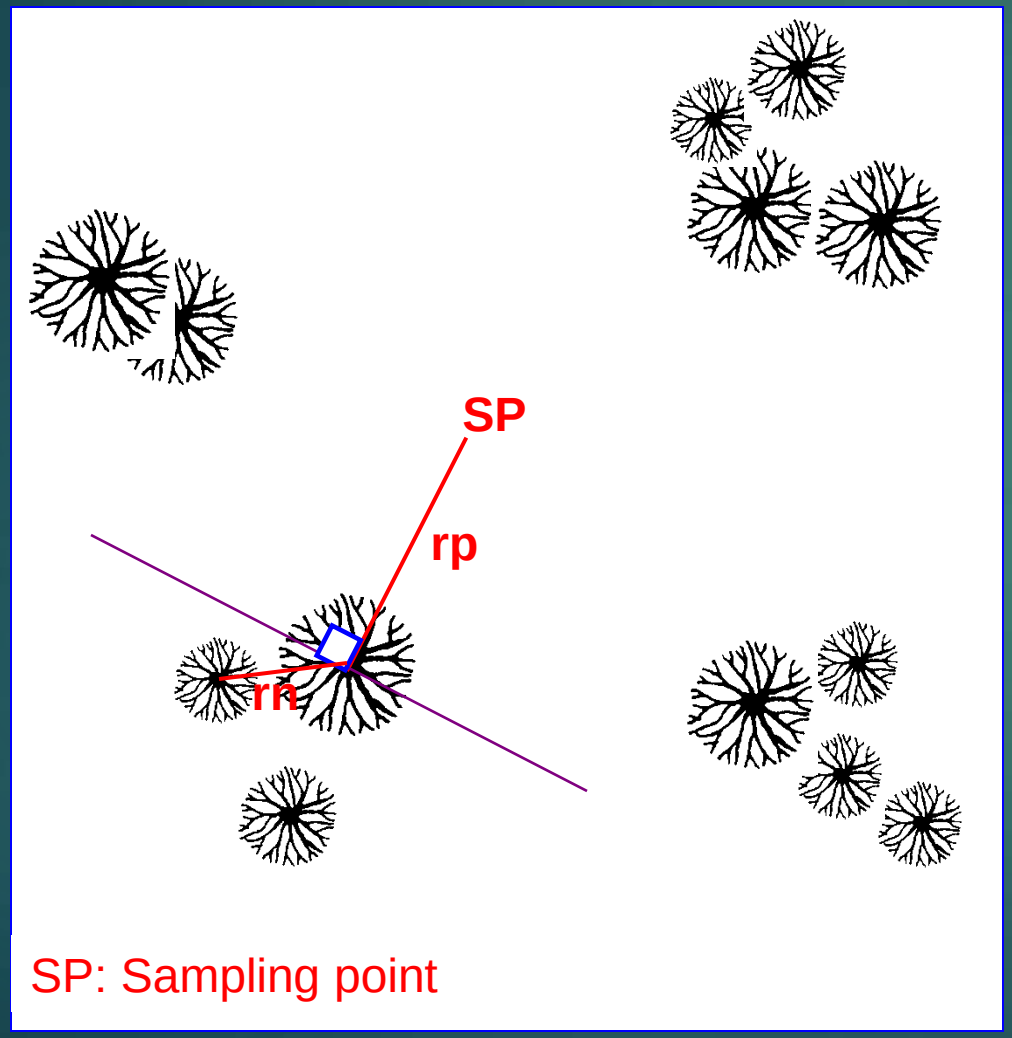


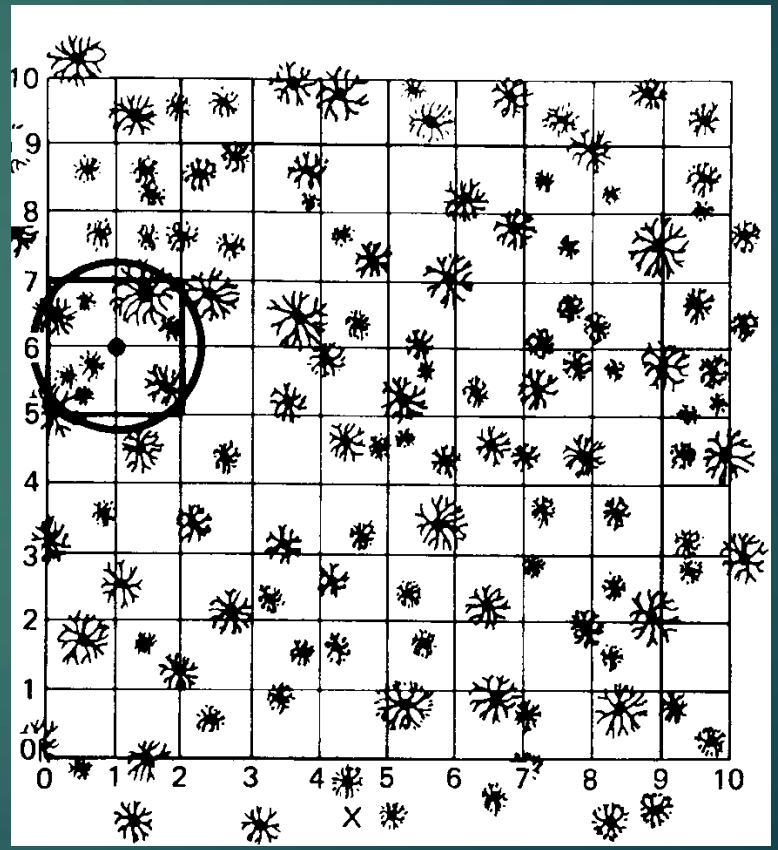
زوج‌هاي تصادفي (Random Pairs)



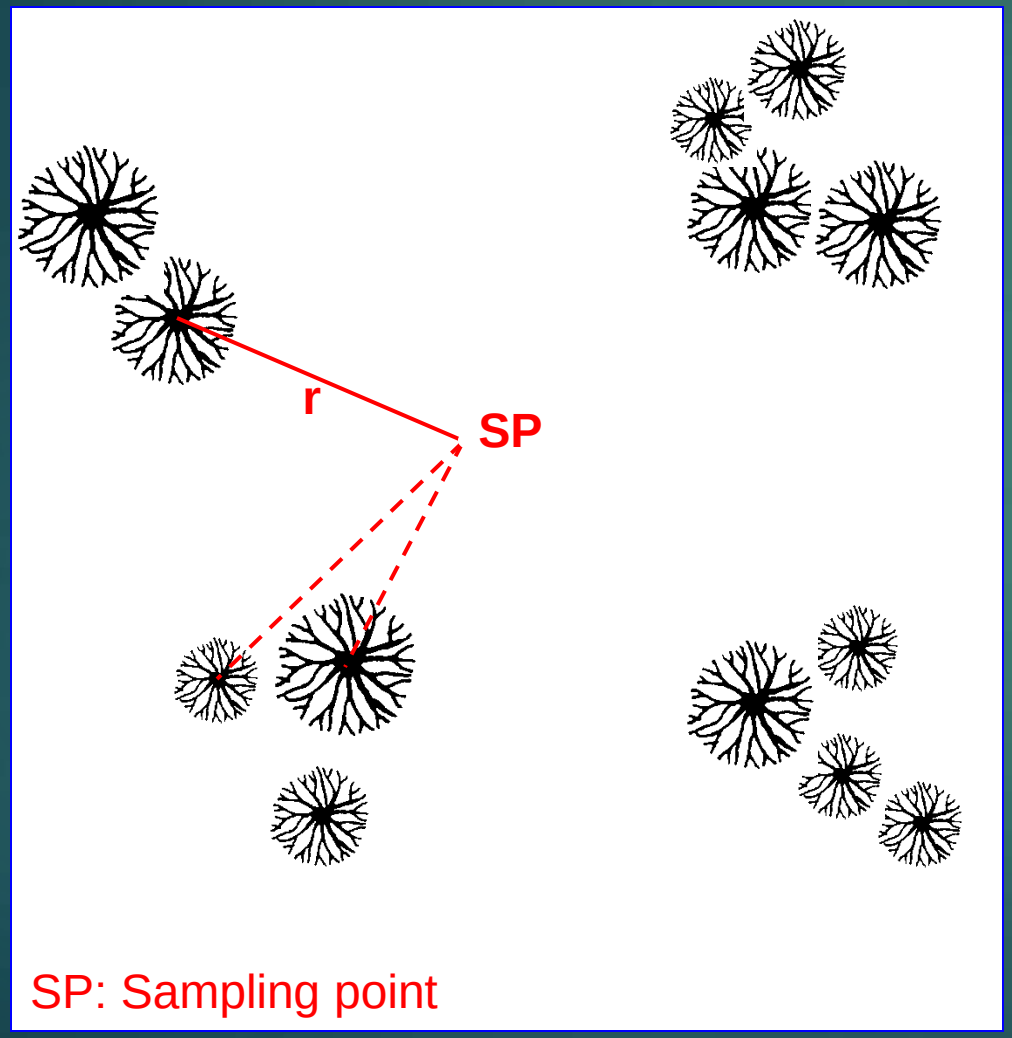


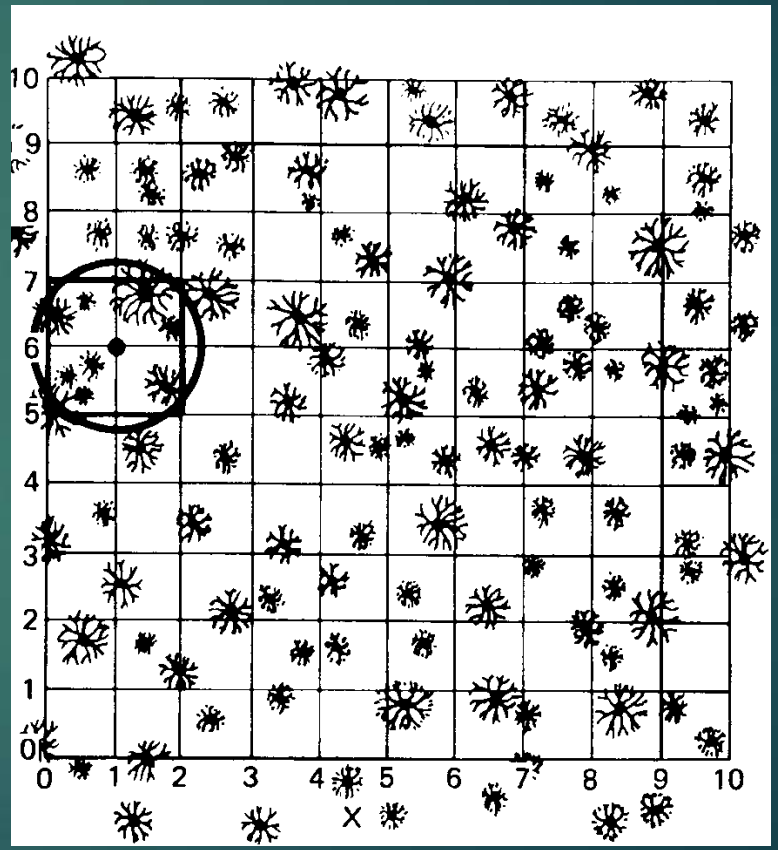
مربع تي (T-square)



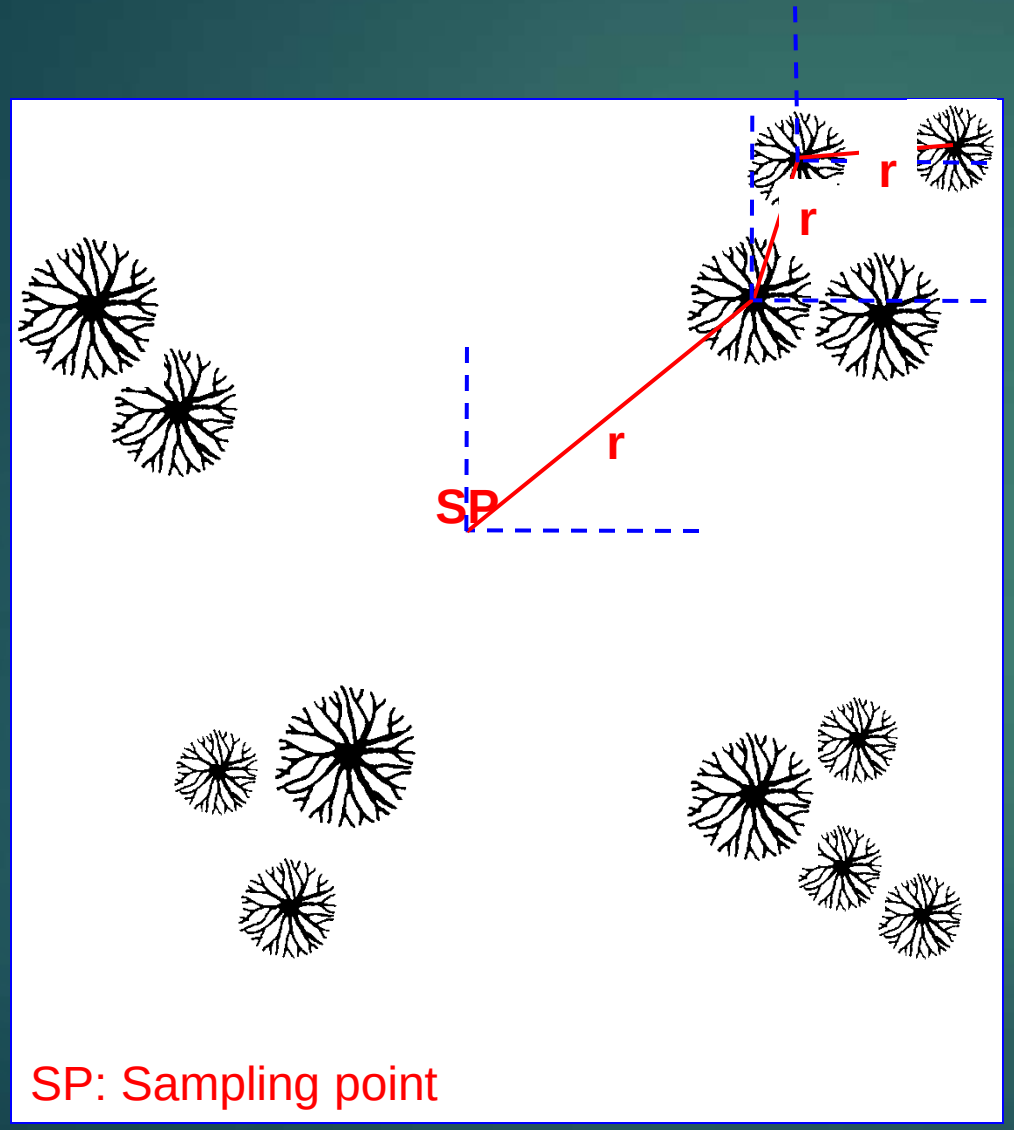


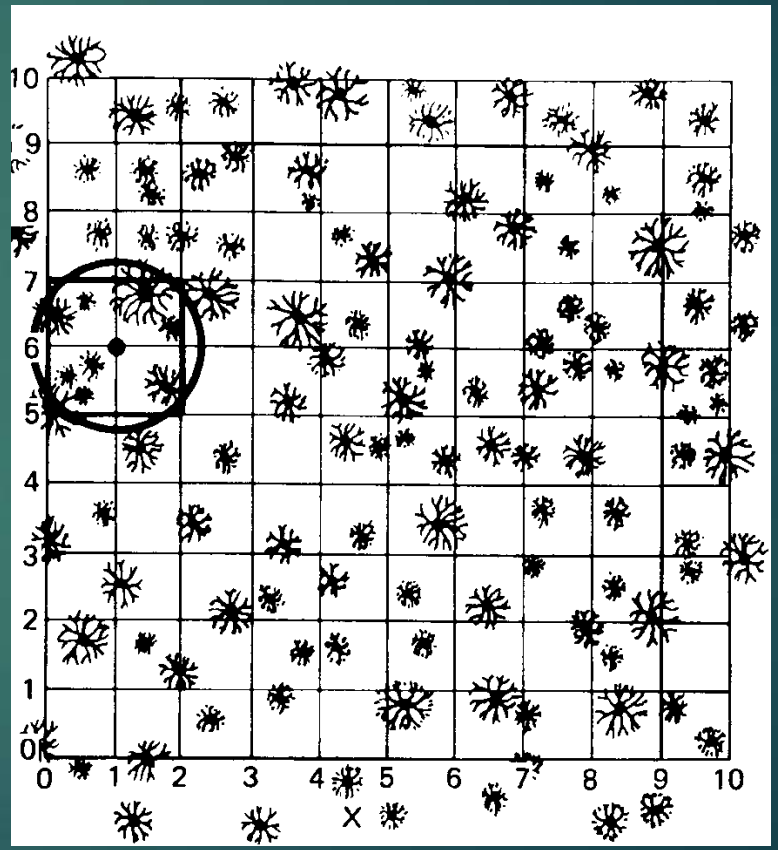
فاصله مرتب (Ordered Distance)



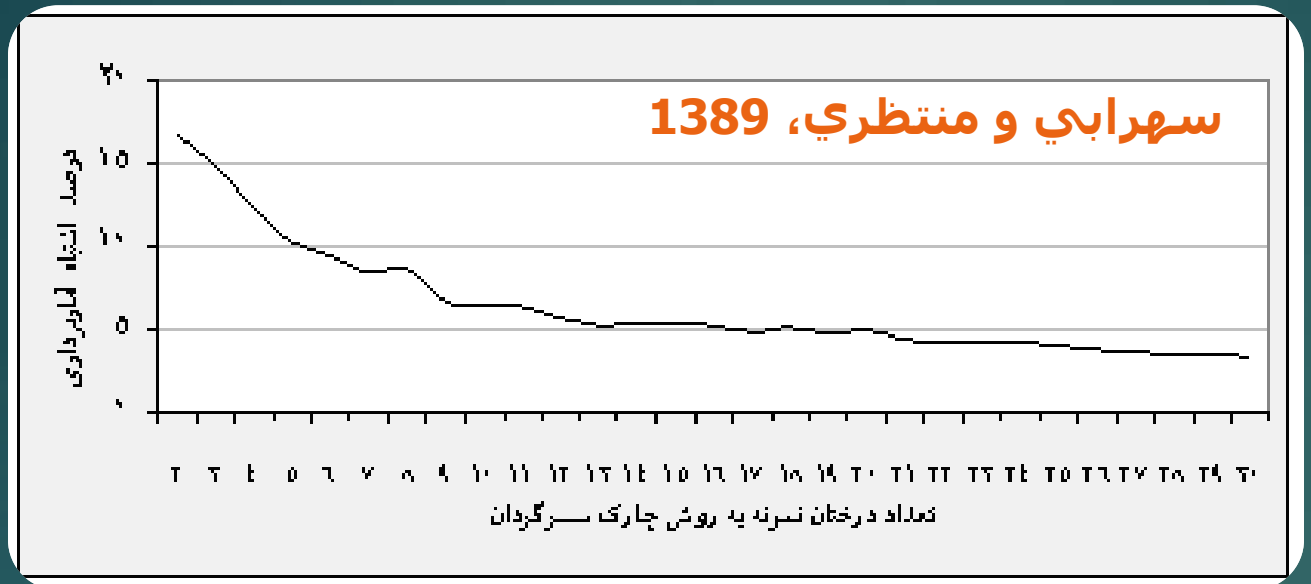


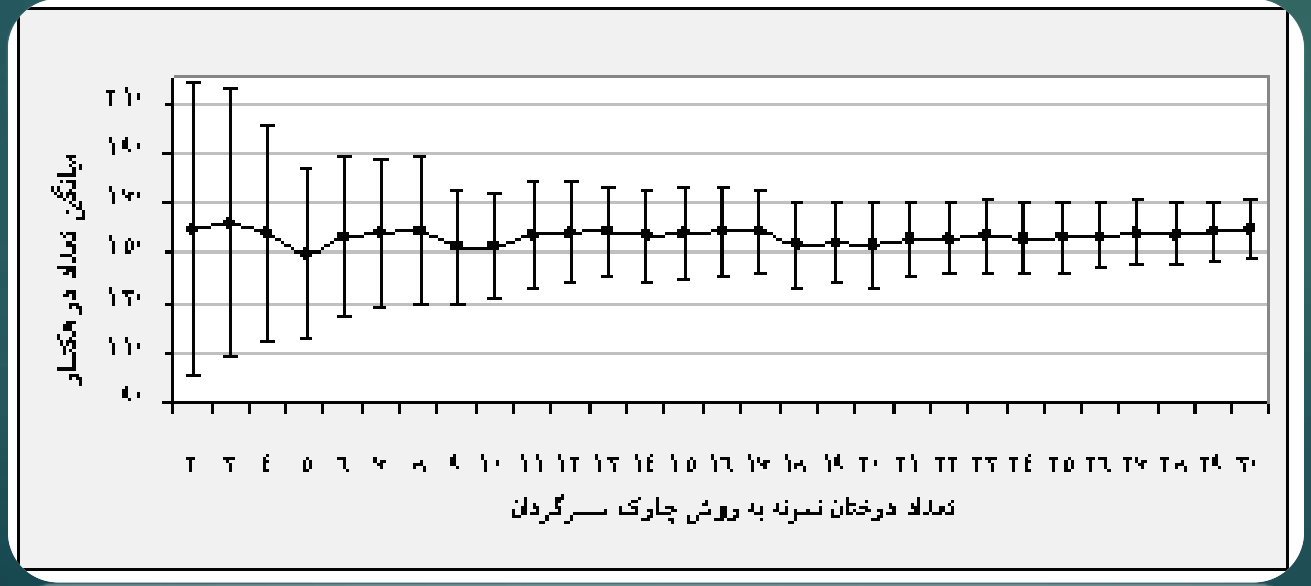
چارك سرگردان (Wandering Quarter)





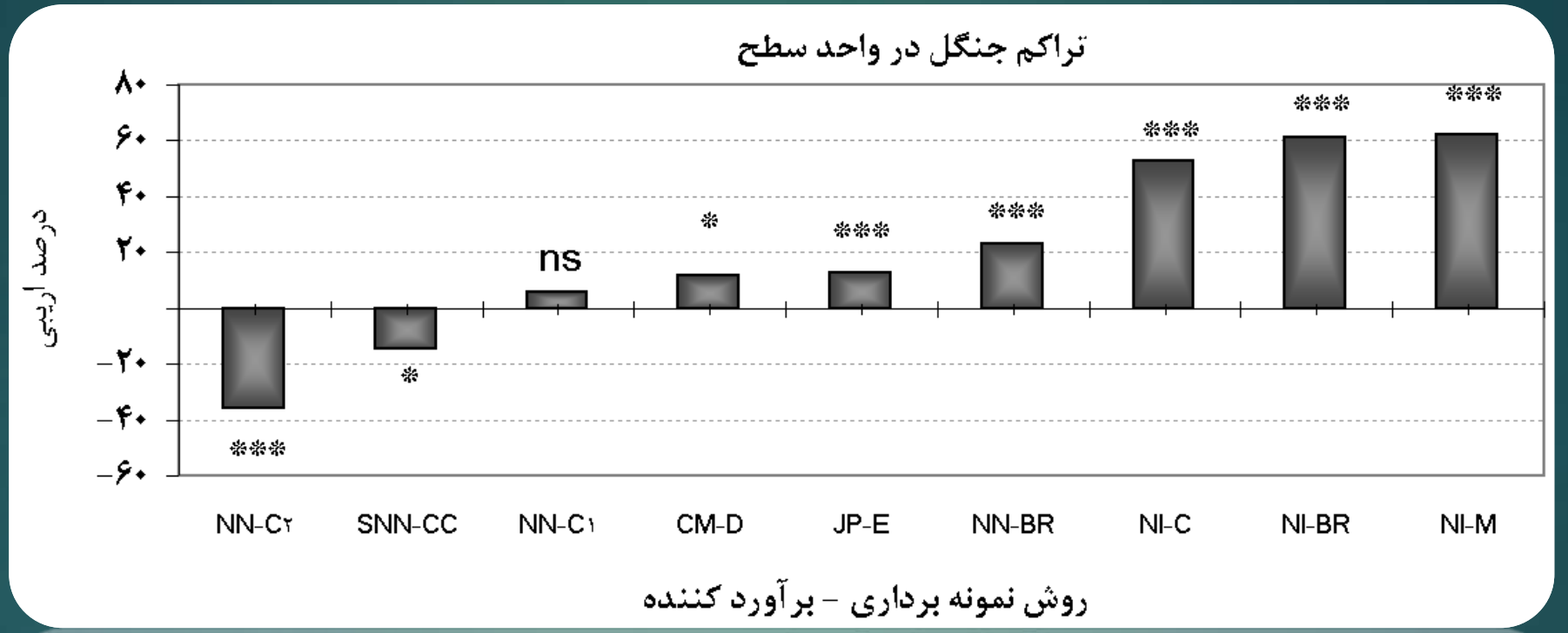
بهينه يابي چارك سرگردان



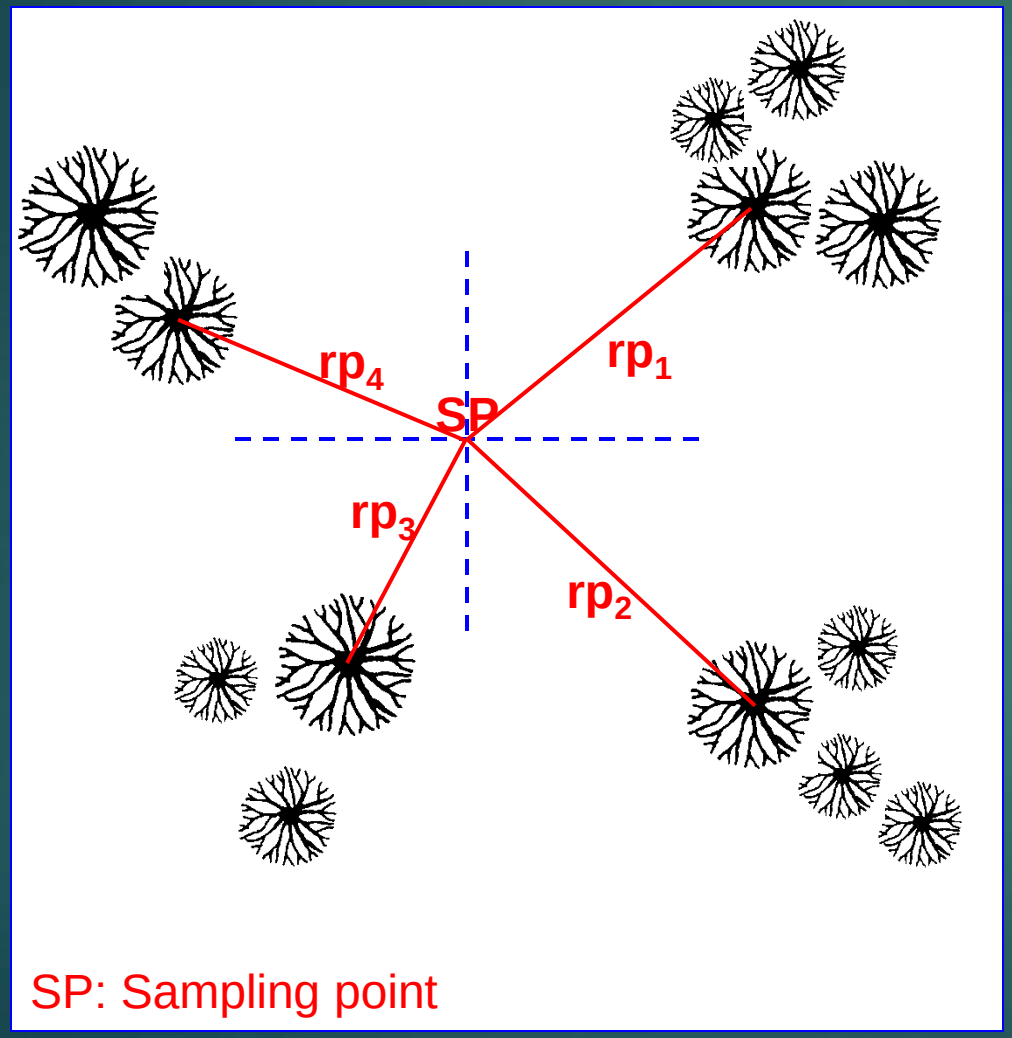


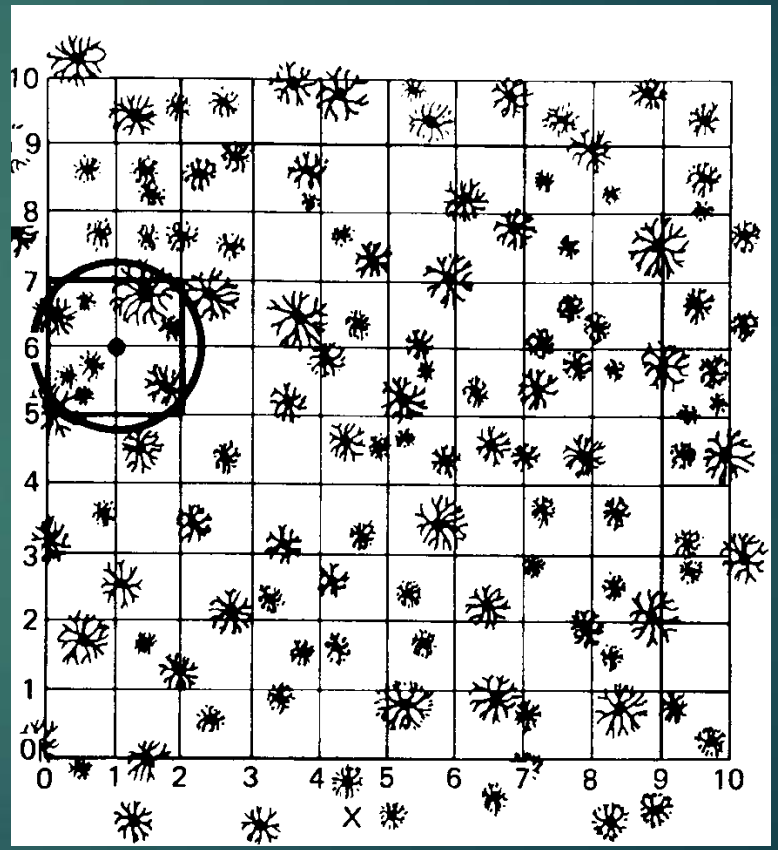
اريبي روش‌ها

عسكري، زبيري و سهرابي ، 1391



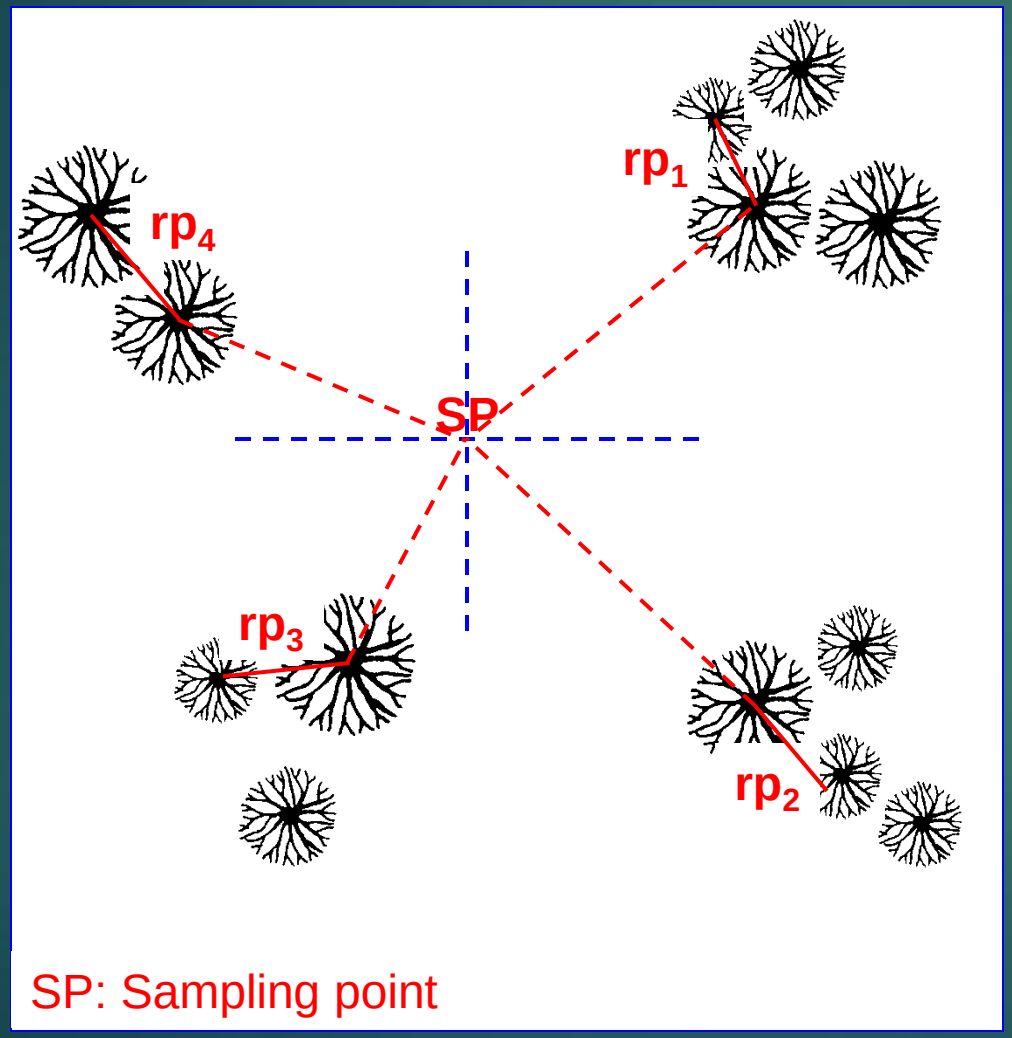
چارك با نقطه مركزي (Point Centered Quarter)

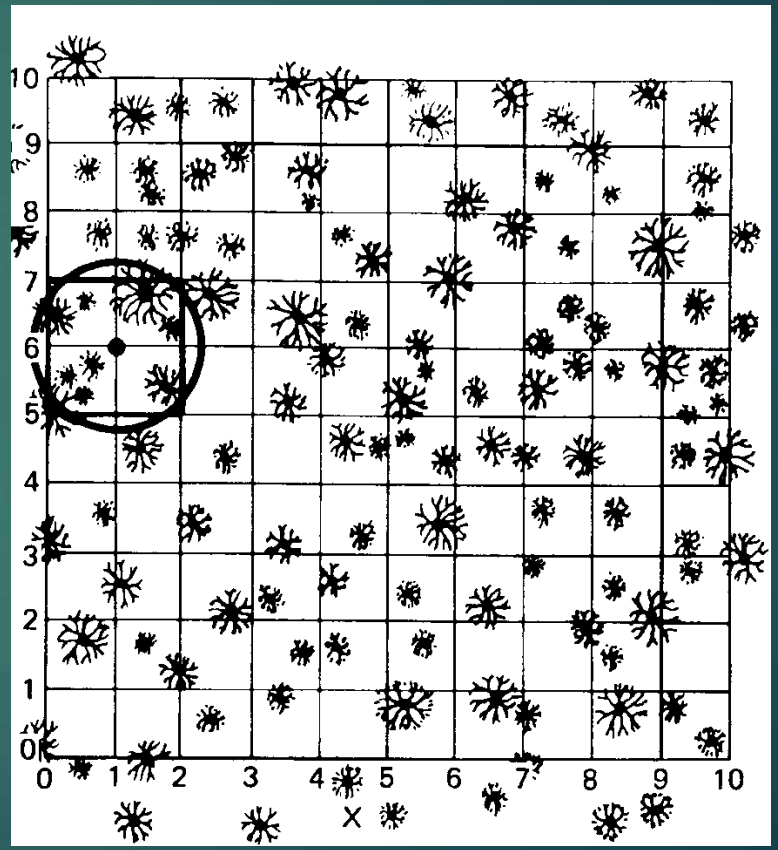




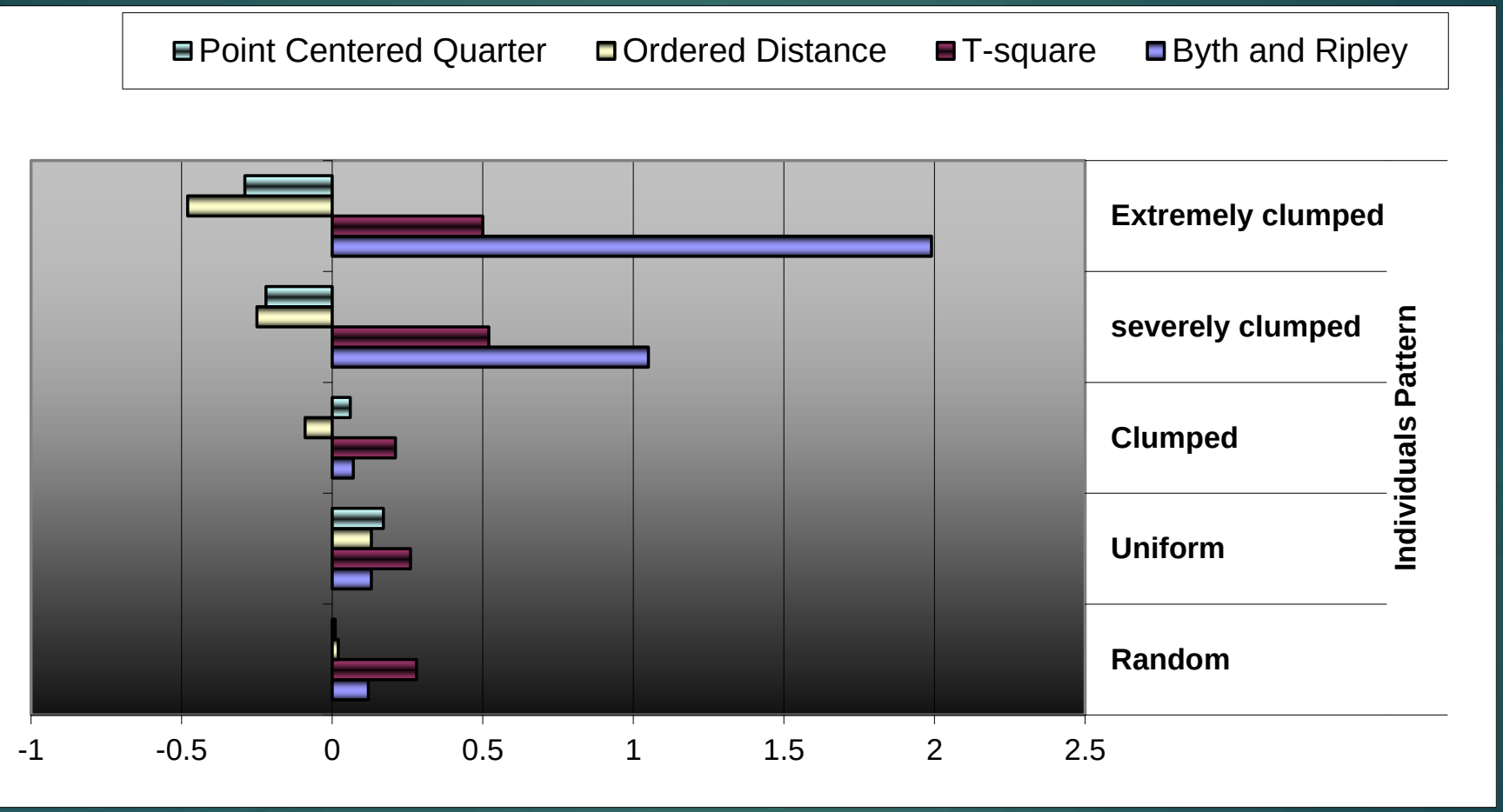
نزديك‌ترين همسايه چاركي (Quartered neighbor method)

Xunzhi ZHU, Jintun ZHANG, 2009. Quartered neighbor method: A new distance method for density estimation. Front. Biol. China, 4(4): 574–578





اريبي روش‌ها

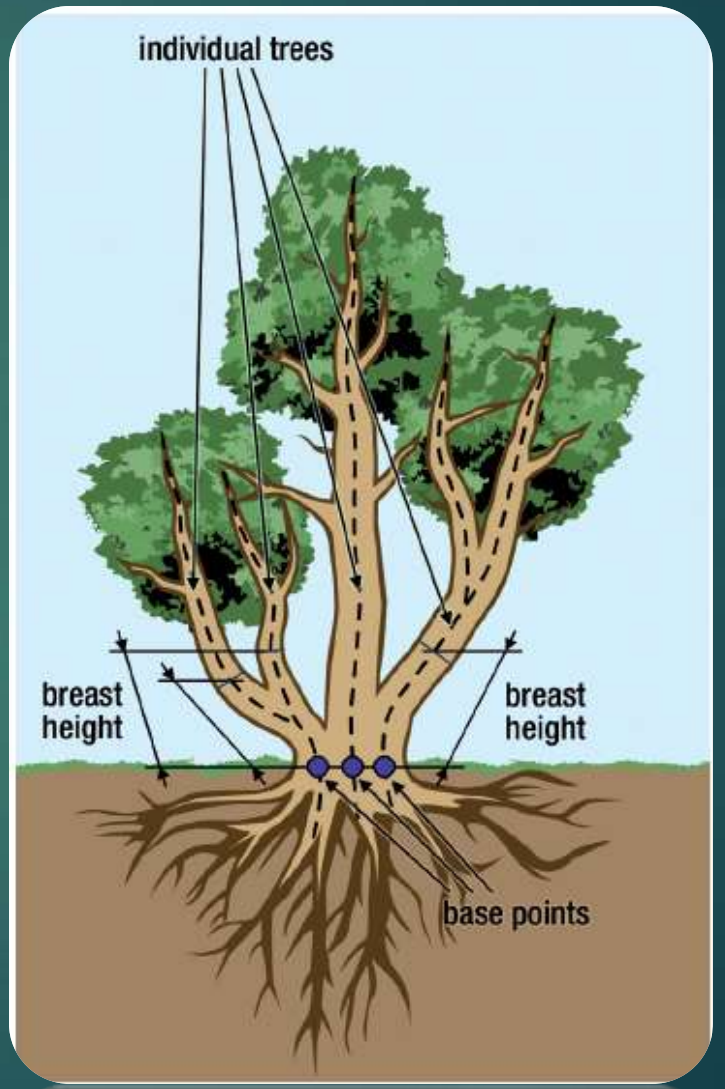


5000 times for four samples sizes: Engeman et al., 1994

چارك با نقطه مركزي اصلاح شده (PCQ+)

Dahdouh-Guebas and Koedam. 2006. *Empirical estimate of the reliability of the use of the Point-Centered Quarter Method (PCQM): Solutions to ambiguous field situations and description of the PCQM+ protocol*, Forest Ecology and Management 228: 1–18

Problems for Multi-stemmed trees (MST)



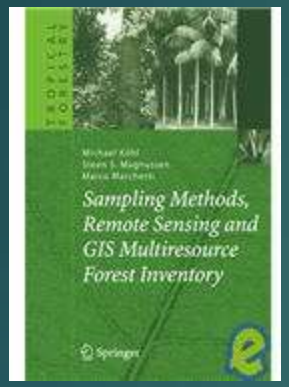
منابع این جلسه

Ecological Methodology
***Krebs*, 1998**
Benjamin Cummings.
Chapter 5: Estimating abundance

روش‌هاي نمونه برداري فاصله‌اي در جنگل
حيدري، 1387
انتشارات دانشگاه رازي

صفحه 1 تا 60

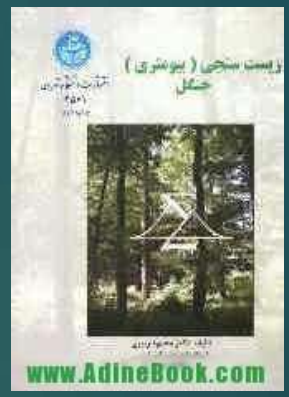
منابع



Sampling Methods, Remote Sensing And GIS Multiresource Forest Inventory

***Kohl et al*. 2006**

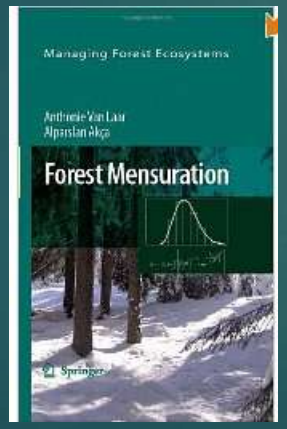
Springer.



زيست سنجي جنگل

زبيري، 1386

انتشارات دانشگاه تهران، چاپ دوم.

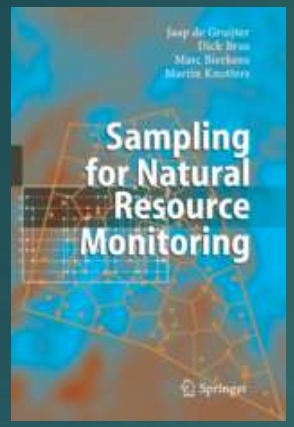


Forest Mensuration

***Laar and Akca*, 2007**

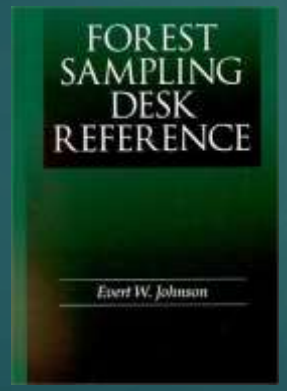
Springer; 2nd edition.

منابع



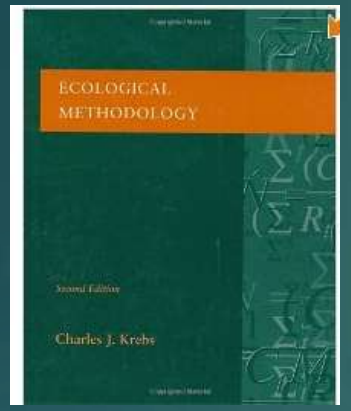
Sampling for Natural Resource Monitoring
Gruijter **et al. 2006**
Springer.

روش‌هاي نمونه برداري فاصله‌اي در جنگل
حيدري، 1387
انتشارات دانشگاه رازي

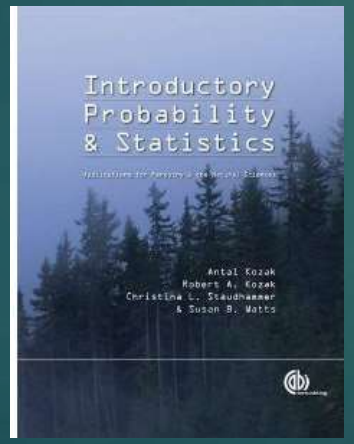


Forest Sampling Desk Reference
Johnson**, 2000**
CRC Press.

منابع



Ecological Methodology
***Krebs*, 1998**
Benjamin Cummings.



Introductory probability and statistics
Kozak, et al., 2008
CABI Press.

برآورد کننده نزدیك ترین فرد

Bythe and Ripley (1980) *Estimators* :

$$\hat{N}_1 = \frac{n}{\pi \sum r p_i^2}$$

$$\text{Define: } \hat{y} = \frac{1}{\hat{N}}$$

$$S_{\hat{y}}^2 = \frac{\hat{y}^2}{n}$$

برآورد کننده نزدیك‌ترین همسایه

Bythe and Ripley (1980) *Estimators* :

$$\hat{N}_2 = \frac{n}{\pi \sum rn_i^2}$$

sample	rn	rn^2
1	12.5	156.0
2	1.5	2.1
3	22.2	490.6
4	2.0	4.2
5	1.5	2.3
6	0.8	0.6
7	3.2	9.9
8	1.9	3.4
9	2.8	7.8
10	1.5	2.3
Sum(sq)		679.2
n		10
Nhat (tree/m.sq)		0.00469
		N2
N hat (tree/ha)		**46.9**

مثال نزديك‌ترين همسايه

$$\text{Define: } \hat{y} = \frac{1}{\hat{N}}$$

$$S^2_{\hat{y}} = \frac{\hat{y}^2}{n}$$

	N2
Nhat (tree/m^2)	0.00469
y hat	213.282
S^2 y hat	4548.905
SE	21.328
t (9, 0.025)	2.69
E(y hat)	57.3

برآورد کننده نزدیك ترین فرد و نزدیك‌ترین همسایه

Diggle (1975) Estimator:

$$\hat{N}_3 = \sqrt{\hat{N}_1 \hat{N}_2}$$

N hat (tree/ha)	N1	N2	N3
	41.6	46.9	44.2

	N3
N hat (tree/m^2)	0.00442
y hat	226.445
S^2 y hat	5127.745
SE	22.645
t (9, 0.025)	2.69
E(y hat)	60.8

برآورد کننده دومین نزدیك‌ترین همسایه

Cottam and Curtis (1956) Estimator:

$$\hat{N} = \frac{n}{2.778 \times \overline{rm}^2}$$

$$\overline{rm} = \frac{\sum rm_i}{n}$$

Sample	SNN
1	1.4
2	3.7
3	6.5
4	6.4
5	8.3
mean (rm)	5.2
N hat (tree/m²)	0.01305
N hat (tree/ha)	1305.0

برآورد کننده روش مرکب

Batcheler (1971) Estimator:

$$\log \hat{N} = \log(d) - \left(0.01416 - 0.1613 \frac{\sum rp_i}{\sum rm_i} \right)$$

$$d = \frac{p}{\pi \left[\sum_{i=1}^{n \text{ for } rp<R} rp_i^2 + (n-p)R^2 \right]}$$

$$R = Md(rp)$$

$$p = Count(rp_i < R)$$

مثال روش مرکب

sample	rp	rn
1	11.2	12.5
2	10.2	1.5
3	17.8	22.2
4	1.2	2.0
5	7.3	1.5
6	1.4	0.8
7	3.7	3.2
8	6.5	1.9
9	6.4	2.8
10	8.3	1.5
md(rp)=R	6.9	
p	5.0	
n	10.0	
sum(rp<R)^2	99.0965	
(n-p)R^2	236.32813	
d	0.00475	

$$d = \frac{p}{\pi\left[\sum_{i=1}^{n\ for\ rp<R} rp_i^2 + (n-p)R^2\right]}$$

$$\log \hat{N} = \log(d) - \left(0.01416 - 0.1613\frac{\sum rp_i}{\sum rn_i}\right)$$

log(d)	-2.324
Log(N hat)	-2.226
N hat (tree/m^2)	**0.006**
N hat (tree/ha)	**59.47**

برآورد کننده نقاط متصل

Engeman et. al (1994) Estimator:

$$\hat{N} = \frac{\hat{N}_1 + \hat{N}_2 + \hat{N}_3}{3}$$

while:

$$\hat{N}_1 = \frac{1}{4\overline{rp}^2}$$

$$\hat{N}_2 = \frac{1}{2.778\overline{rn}^2}$$

$$\hat{N}_3 = \frac{1}{2.778\overline{rm}^2}$$

مثال نقاط متصل

$$\hat{N} = \frac{\hat{N}_1 + \hat{N}_2 + \hat{N}_3}{3}$$

$while:$

$$\hat{N}_1 = \frac{1}{4\overline{rp}^2}$$

$$\hat{N}_2 = \frac{1}{2.778\overline{rn}^2}$$

$$\hat{N}_3 = \frac{1}{2.778\overline{rm}^2}$$

sample	rp	rn	rm
1	11.2	12.5	1.4
2	10.2	1.5	3.7
3	17.8	22.2	6.5
4	1.2	2.0	6.4
5	7.3	1.5	8.3
6	1.4	0.8	6.1
7	3.7	3.2	2.9
8	6.5	1.9	6.5
9	6.4	2.8	2.3
10	8.3	1.5	1.9

N1	N2	N3
0.00459	0.01455	0.01708

N hat (tree/m²) **0.01207**

N hat (tree/ha) **120.7**

برآورد کننده‌هاي زوج‌هاي تصادفي

Cottam and Curtis (1956) Estimator:

$$\overline{rn} = \frac{\sum rn_i}{n}$$

$$\hat{N} = \frac{1}{(0.8 \times \overline{rn})^2}$$

sample	rn
1	10.9
2	16.0
3	10.2
4	21.2
5	16.1
6	14.1
7	17.0
8	14.2
9	14.2
10	13.0
11	15.0
12	30.1

mean (rn)	15.8
N hat (tree/m^2)	0.00619
N hat (tree/ha)	**61.9**

برآورد کننده مربع تي

$$\hat{N} = 2\left(\frac{n}{\pi \sum rn_i^2}\right)$$

Byth (1982) Estimator:

$$\hat{N}_T = \frac{n^2}{2.282 \sum rp_i \sum rn_i}$$

مثال مربع تي

Sample	rp	rp^2	rn	for Nt	
1	12.6	158.76	8.7	sum(rp)	86.5
2	9.3	86.49	16.4	sum(rn)	87.1
3	7.5	56.25	9.3	N hat	0.00380
4	16.2	262.44	12.6	N hat	38.0
5	8.8	77.44	3.5		
6	10.1	102.01	11.2		
7	6.2	38.44	13.6		
8	1.5	2.25	9.1		
9	14.3	204.49	2.7		
sum.sq		988.57			
N hat (tree/m.sq)		0.0058			
N hat (tree/ha)		**58.0**			

$$\hat{N}_T = \frac{n^2}{2.282 \sum rp_i \sum rn_i}$$

$$\hat{N} = 2\left(\frac{n}{\pi \sum rn_i^2}\right)$$

برآورد کننده فاصله مرتب

Morisita (1957), Pollard (1971) Estimator:

$$\hat{N} = \frac{3n-1}{\pi \sum r_i^2}$$

$$Varianc(\hat{N}) = \frac{\hat{N}^2}{3n-2}$$

$$\text{Lower confidence limit for } \hat{N} = \left(\frac{\sqrt{12n-1} - 1.96}{\sqrt{4\pi \sum r_i^2}} \right)^2$$

$$\text{Upper confidence limit for } \hat{N} = \left(\frac{\sqrt{12n-1} + 1.96}{\sqrt{4\pi \sum r_i^2}} \right)^2$$

مثال فاصله مرتب

$$\hat{N} = \frac{3n-1}{\pi \sum r_i^2}$$

$$Varianc(\hat{N}) = \frac{\hat{N}^2}{3n-2}$$

$$CI(\pm) = \left(\frac{\sqrt{12n-1} \pm 1.96}{\sqrt{4\pi \sum r_i^2}} \right)^2$$

sample	r	r^2
1	11.2	125.44
2	10.2	104.04
3	17.8	316.84
4	1.2	1.44
5	7.3	53.29
6	1.4	1.96
7	3.7	13.69
8	6.5	42.25
9	6.4	40.96
10	8.3	68.89
Sum		768.8
N hat (m2)		0.012013
N hat (ha)		120.131
Variance		515.4089

برآورد کننده PCQ

Pollard (1971) Estimator:

$$\hat{N} = \frac{4(4n-1)}{\pi \sum rp_{ij}^2}$$

$$Varianc(\hat{N}) = \frac{\hat{N}^2}{4n-2}$$

$$SE = \sqrt{\frac{variance(\hat{N})}{4n}}$$

$$\text{Lower confidence limit for } \hat{N} = \left(\frac{\sqrt{16n-1}-1.96}{\sqrt{\pi \sum rp_{ij}^2}} \right)^2$$

$$\text{Upper confidence limit for } \hat{N} = \left(\frac{\sqrt{16n-1}+1.96}{\sqrt{\pi \sum rp_{ij}^2}} \right)^2$$

مثال PCQ

$$\hat{N} = \frac{4(4n-1)}{\pi \sum rp_{ij}^2}$$

$$Varianc(\hat{N}) = \frac{\hat{N}^2}{4n-2}$$

$$SE = \sqrt{\frac{variance(\hat{N})}{4n}}$$

$$CI(\pm) = \left(\frac{\sqrt{16n-1} \pm 1.96}{\sqrt{\pi \sum rp_{ij}^2}} \right)^2$$

sample	rp1	rp2	rp3	rp4
1	11.2	5.1	10.78	8.3
2	10.2	12.04	8.17	20.36
3	17.75	7.17	3.01	7.57
4	1.18	2.82	5.73	12.8
5	7.29	5.84	14.14	5.6
6	1.4	10.44	8.75	8.14
7	3.7	2.47	6.24	4.95
8	6.46	13.16	6.49	11.64
9	6.35	3.03	2.78	18.05
10	8.3	10.99	4.9	0.67
SS	3350.929			
N hat	0.01483			
N hat (ha)	148.3			

برآورد کننده چارك سرگردان

Catana (1950) Estimator:

$$\hat{N} = \frac{1}{\bar{r}}$$

$$\bar{r} = \frac{\sum r_i}{n}$$

sample	r
1	11.2
2	10.2
3	17.8
4	1.2
5	7.3
6	1.4
7	3.7
8	6.5
9	6.4
10	8.3
mean	7.4
N hat	0.0183
N hat (ha)	183.5